Quiz — 1.4.1 Data Types

Name: ____ Date: _ ***Mark:*** ___ / 30

OCR H446 · Component 01 · 1.4.1 — show all working

Section A — Multiple choice (8 marks)

1. How many bits are represented by a single hexadecimal digit? A. 2 B. 4 C. 8 D. 16 Your answer: ☐

2. What is the denary value of the unsigned binary number `00101101`? A. 43 B. 44 C. 45 D. 47 Your answer: ☐

3. In an 8-bit two's complement number, what is the place value of the most significant bit? A. +128 B. −128 C. +127 D. −127 Your answer: ☐

4. Which operation has the effect of multiplying an unsigned binary number by 4? A. Logical shift right 2 places B. Logical shift left 2 places C. Arithmetic shift right 2 places D. Logical shift left 4 places Your answer: ☐

5. In a floating-point number, which part determines the **precision**? A. The exponent B. The sign bit C. The mantissa D. The base Your answer: ☐

6. A correctly normalised **positive** floating-point mantissa always begins with which bit pattern? A. `1.0…` B. `0.0…` C. `1.1…` D. `0.1…` Your answer: ☐

7. The ASCII code for `'A'` is 65. What is the ASCII code for `'D'`? A. 66 B. 67 C. 68 D. 69 Your answer: ☐

8. Which statement about Unicode is correct? A. Unicode can only represent the English alphabet B. ASCII is a subset of Unicode, so Unicode is backward-compatible C. Unicode always uses exactly 7 bits per character D. Unicode cannot represent emoji Your answer: ☐

Section B — Calculations & short answer (17 marks)

9. Convert the denary number **217** to 8-bit unsigned binary. **[2]**

Working space:

10. Convert the binary number `10110110` to hexadecimal. **[2]**

Working space:

11. Convert the hexadecimal number `B7` to denary. **[2]**

Working space:

12. Using 8-bit two's complement, calculate **45 − 28** by adding the two's complement of 28. Show the binary working and state what happens to the final carry. **[4]**

Working space:

13. Normalise the following 8-bit signed floating-point number. The mantissa is `0.0110100` (one sign bit then the fraction) and the exponent is `0011` (4-bit two's complement). Give the normalised mantissa and the new exponent, and explain the rule used. **[4]**

Working space:

14. The 8-bit unsigned binary number `00010110` is shifted using a **logical left shift of 2 places**. Give the result and state the arithmetic effect. **[3]**

Working space:

Section C — Exam-style question (5 marks)

15. A programmer stores temperatures using floating-point representation with a fixed total number of bits.

(a) Explain the trade-off between allocating more bits to the mantissa versus more bits to the exponent. **[2]**

(b) The programmer says "I'll just use hexadecimal to store the numbers so they take less memory." Explain whether using hexadecimal reduces the amount of memory used, and give one genuine reason programmers use hexadecimal. **[3]**

Working space: